

Supplementary Material: Gate-Based Quantum Optimization for Discrete Process Design: A Repair-Aware Benchmark

Yirang Park^a and David E. Bernal Neira^a

^a Davidson School of Chemical Engineering, Purdue University, West Lafayette, IN 47907, USA

ABSTRACT

This supplement provides four sections of supporting material for the main manuscript. Section 1 describes the computational environment and data availability. Section 2 reports numerical validation checks supporting the manuscript claims, including the surrogate rank audit. Section 3 derives the exact auxiliary-repair decomposition that reduces per-sample repair to 34 comparisons. Section 4 documents algorithm parameters and hardware execution metadata for the variational and IBM Quantum pilot experiments.

Keywords: supplementary material, quantum optimization, process design, reproducibility

COMPUTATIONAL ENVIRONMENT AND DATA AVAILABILITY

All scripts, data files, and circuit records for this case study are archived at <https://github.com/SECQUOIA/pse-quantum-discrete-optimization> under `case-studies/ds-mfg-qubo-qiskitopt/`. The repository environment can be verified with `julia --project=. scripts/smoke_test.jl`, which imports all packages, checks required artifacts, and validates cached summary values against recorded checksums. Table 1 lists the pinned software versions used to produce the results in the main manuscript.

Table 1: Pinned software environment for the DS-MFG case study. Version specifications are recorded in `case-studies/ds-mfg-qubo-qiskitopt/Project.toml`, `Manifest.toml`, and `CondaPkg.toml`.

Layer	Component and version	Configuration file
Julia optimization wrapper	QiskitOpt.jl v0.7.0.	Project.toml;Manifest.toml.
Julia QUBO stack	QUBODrivers.jl v0.6.1, QUBOTools.jl v0.13.0, QUBO.jl v0.6.0.	Project.toml;Manifest.toml.
Python bridge	Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3.	CondaPkg.toml.
Classical solver	Gurobi 13.0.2 (PoolSearchMode=2, PoolSolutions=36).	scripts/run_classical_baselines.jl.

NUMERICAL VALIDATION

Table 2: Numerical validation checks supporting the manuscript claims.

Check	Value	Anchor
Exact repaired optimum	$F(y^*) = 11.7095$ at flow 1001110100111100011; certified by exact enumeration over all 2^{19} flow assignments.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv.
Gurobi pool bridge	All 36 Gurobi-pool flows match the repaired QUBO score within 3.1×10^{-13} ; Gurobi 13.0.2 rerun returns OPTIMAL, MIPGap=0.0, 36 solutions, exact pool agreement.	ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_solutions.csv.
Quadratic surrogate weakness	Global fit $R^2 \approx 0.999$, but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; Figure 1.
Low-energy surrogate audit	Exact enumeration finds 36 feasible flows. The 50 lowest-cost repaired flows include 14 infeasible flows — the penalty weights are insufficient to push all infeasible flows above all feasible ones. Spearman rank correlation between exact and surrogate ordering in the lowest 1% repaired-flow set is 0.901.	ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; MANUSCRIPT_FINDINGS.tex.
Primary QAOA simulator check	JuliQAOA statevector probability on the reference optimum was 2.63×10^{-3} for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was 2.53×10^{-3} , inside the Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$.	ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv; ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv.

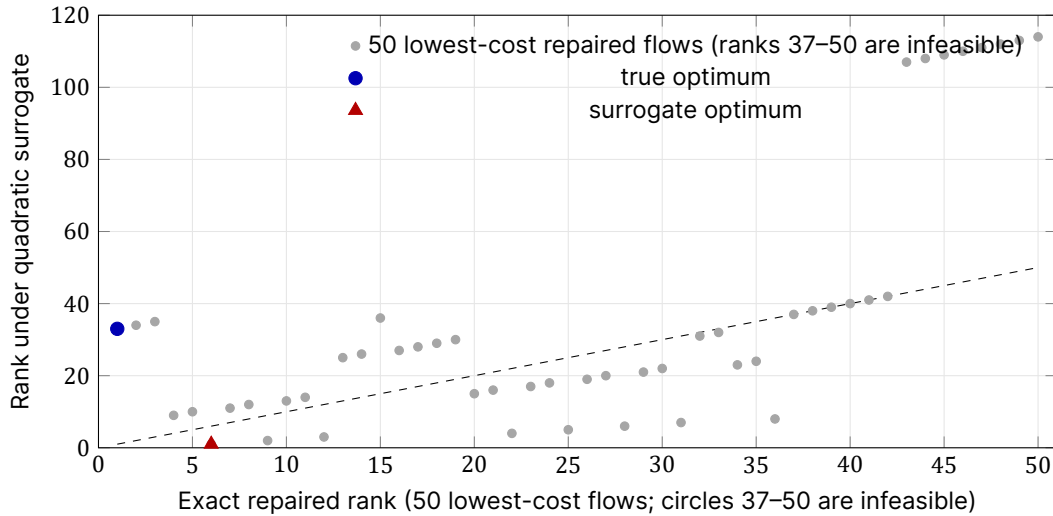


Figure 1. Surrogate rank audit for the 50 lowest-cost repaired flows. Exact enumeration confirms there are exactly 36 feasible flows (ranks 1–36); the remaining 14 shown here are infeasible flows whose penalty values are insufficient to place them above all feasible ones. The surrogate has high global R^2 but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective F , not the surrogate $\hat{F}$.

EXACT AUXILIARY REPAIR: COMPONENT DECOMPOSITION

The QUBO objective $E_{\text{QUBO}}(y, z)$ acts on 36 binary variables: the 19 flow (design) variables $y = (y_1, \dots, y_{19})$ and 17 binary auxiliary (slack) variables $z = (z_1, \dots, z_{17})$ introduced by the penalty reformulation. A sampled 36-bit string is not directly interpretable as a flowsheet because the auxiliary bits are bookkeeping for the inequality constraints of the original design problem, not design decisions. After projecting a sample onto the 19 flow bits y , the repair step finds the auxiliary assignment that minimizes the QUBO energy for that flow:

$$z^*(y) = \arg \min_{z \in \{0,1\}^{17}} E_{\text{QUBO}}(y, z). \quad (1)$$

Brute-force enumeration over all $2^{17} = 131,072$ auxiliary assignments is tractable for a single call but unnecessarily costly when repair is applied to every sampled state across all workflows. Once y is fixed, every term in E_{QUBO} that involves an auxiliary variable z_k is either (i) a linear term $c_k z_k$, (ii) an auxiliary-auxiliary coupling $c_{k\ell} z_k z_\ell$, or (iii) a flow-auxiliary coupling $c_{ik} y_i z_k$ (a constant given y). Consider the *auxiliary coupling graph*: nodes are the 17 auxiliary variables, and an edge connects z_k and z_ℓ whenever their quadratic coefficient $c_{k\ell} \neq 0$. For this QUBO the graph has 13 connected components: 9 isolated nodes (one auxiliary bit each) and 4 edges (two auxiliary bits each). Because there are no edges between components, fixing y makes the components conditionally independent, and (1) decomposes as

$$\min_z E_{\text{QUBO}}(y, z) = \text{const}(y) + \sum_{c=1}^{13} \min_{z^{(c)}} E^{(c)}(y, z^{(c)}), \quad (2)$$

where $z^{(c)}$ are the auxiliary bits in component c and $E^{(c)}$ contains only the terms that involve those bits (with fixed y values substituted as constants). Each sub-problem is solved by enumerating $2^1 = 2$ or $2^2 = 4$ assignments and selecting the minimum energy; the total work is at most $9 \times 2 + 4 \times 4 = 34$ comparisons per repair, a factor of $\sim 3,800\times$ smaller than the brute-force bound. This procedure is implemented in `scripts/run_classical_baselines.jl`; the component structure is read from `ds_mfg_reduced_flow_objective/auxiliary_components.csv`.

Physical meaning of the components. Table 3 lists all 13 components with their auxiliary variable indices, the flow variables whose fixed values enter each local energy, and a brief description. Singleton components (1–5, 10–13) correspond to individual inequality constraints in the penalty QUBO: the single binary slack takes the value (0 or 1) that satisfies the inequality, which for feasible

flow assignments is the constraint residual. Pair components (6–9) each arise from the product-linearization of the Route B activation rule ($f_{06} = 1$ iff Reactor 1 is PFR or CSTR and Reactor 2 is Batch). Linearizing a product of two binary variables requires up to three inequality constraints; the slack for such a constraint can take values in $\{0, 1, 2\}$, needing two binary bits. The four pair components together represent the complete linearization of this one coupling rule across all Reactor 1/Reactor 2 type combinations.

Table 3: Independent auxiliary components for the DS-MFG QUBO repair (source: `ds_mfg_reduced_flow_objective/auxiliary_components.csv`). Variable indices 1–19 are the 19 flow variables y ; indices 20–36 are the 17 binary auxiliary (slack) variables z . *Flow neighbors* are the y -indices that appear in the local energy $E^{(c)}$. No coupling term links auxiliaries across components, so each sub-problem is solved independently by enumeration.

#	Aux. idx.	n_{aux}	Flow neighbors (y idx. and f -arc)	$2^{n_{\text{aux}}}$	Role
1	20	1	17 (f_{16} , Batch U/S), 18 (f_{17} , cont. exit)	2	Batch/cont. downstream exclusion
2	21	1	2 (f_{01} , PFR Rxn 1), 3 (f_{02} , CSTR Rxn 1), 4 (f_{03} , Batch Rxn 1)	2	Reactor 1 at-most-one
3	22	1	6 (f_{05} , Route A), 7 (f_{06} , Route B)	2	Route at-most-one
4	23	1	9 (f_{08} , PFR Rxn 2), 10 (f_{09} , CSTR Rxn 2), 11 (f_{10} , Batch Rxn 2)	2	Reactor 2 at-most-one
5	24	1	14 (f_{13} , Cont. I), 15 (f_{14} , Cont. II), 16 (f_{15} , Cont. III), 17 (f_{16} , Batch U/S)	2	Downstream at-most-one
6	25, 26	2	2 (f_{01} , PFR Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (PFR Rxn 1 case)
7	27, 28	2	3 (f_{02} , CSTR Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (CSTR Rxn 1 case)
8	29, 30	2	4 (f_{03} , Batch Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (Batch Rxn 1 case)
9	31, 32	2	2 (f_{01} , PFR Rxn 1), 3 (f_{02} , CSTR Rxn 1), 6 (f_{05} , Route A)	4	Route B linearization (Route A side)
10	33	1	6 (f_{05} , Route A), 11 (f_{10} , Batch Rxn 2)	2	Route A / Rxn 2 coupling
11	34	1	14 (f_{13} , Cont. I), 18 (f_{17} , cont. exit)	2	Cont. I flow balance
12	35	1	15 (f_{14} , Cont. II), 18 (f_{17} , cont. exit)	2	Cont. II flow balance
13	36	1	16 (f_{15} , Cont. III), 18 (f_{17} , cont. exit)	2	Cont. III flow balance
9×2 + 4×4 = 34 comparisons per repair vs. $2^{17} = 131,072$ brute force ($\sim 3,800\times$ reduction).					

ALGORITHM PARAMETERS AND HARD- WARE EXECUTION

Table 4: Circuit parameters for the QAOA and VQE workflows reported in Table 1 of the main manuscript.

Workflow	Circuit / ansatz	Objective and optimizer	Parameter source	Validation metric
Energy-targeted QAOA	$p = 5$ QAOA on the 19-flow surrogate.	Minimize $\mathbb{E}_{p_\theta}[\tilde{F}(y)]$ with <code>JuliQAOA find_angles_bh</code> ; seed 91001; 5 basin-hopping iterations.	<code>ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv</code> ; angles transferred beta-then-gamma to <code>QiskitOpt.QAOA</code> .	664 reference hits / 262144 reads; 39661 top-50 reads.
Reduced-surrogate VQE	<code>EfficientSU2</code> ; 152 parameters on the 19-flow surrogate.	COBYLA through <code>QiskitOpt.VQE</code> ; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up.	Sampled distributions and optimized ansatz vectors retained in <code>ds_mfg_vqe_reduced_flow_objective_seed74018_metadata/</code> and <code>ds_mfg_vqe_reduced_flow_objective_selected_metadata/</code> .	Seed 74018: 20 reference hits / 524288 reads; three seeds combined: 28 reference hits / 1572864 reads.

Table 5: Hardware pilot execution metadata retained in the submitted-job archive.

Item	Metadata
Backend and jobs	<code>ibm_fez</code> (156 qubits); backend queried 2026-06-17 20:19 UTC; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained in manifest.
Logical circuit	19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 <code>h</code> , 95 <code>rx</code> , 95 <code>rz</code> , 255 <code>rz</code> , and 19 measurements; scoring reverses Qiskit count keys to $x_1, \dots, x_{19}$.
Hardware outcome	36864 total reads; 6 top-50 reads; 4 Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 ($F = 11.8105$). With zero reference hits, the two-sided 95% Wilson upper bound on the reference-optimum rate is 1.04×10^{-4} . The pilot validates submission and scoring; no noise mitigation was applied.